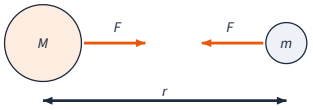
CHAPTER 6 GRAVITATION

Key idea: every mass attracts every other mass; the force falls off as $1/r^2$.

1 NEWTON'S LAW OF GRAVITATION



$$F = \frac{GMm}{r^2}$$

- Always attractive; r between centres
- $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
- Equal and opposite on the two masses

2 FIELD STRENGTH G

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

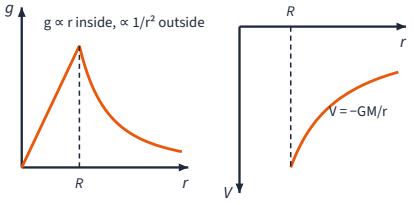
- Force per unit mass, N kg^{-1} (= m s^{-2})
- Surface: $g_0 = \frac{GM}{R^2}$, so $GM = g_0 R^2$
- Height h: $g = \frac{GM}{(R+h)^2}$

3 GRAVITATIONAL POTENTIAL

$$V = -\frac{GM}{r} \qquad U = -\frac{GMm}{r}$$

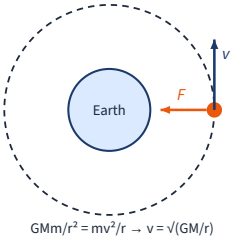
- Work done per unit mass to bring a mass from **infinity**; unit J kg^{-1}
- Zero at infinity, negative elsewhere
- $g = -\frac{dV}{dr}$: field = −(potential gradient)
- Moving m: $W = m\Delta V$

4 INSIDE AND OUTSIDE



- Inside (uniform Earth): $g \propto r$
- Outside: $g \propto 1/r^2$; $V \propto -1/r$
- $g = 0$ at the centre

5 SATELLITE MOTION



$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}} \qquad T^2 = \frac{4\pi^2}{GM} r^3$$

Lower orbit: faster, shorter period.

6 ENERGY OF A SATELLITE

	Expression
Kinetic	$+\frac{GMm}{2r}$
Potential	$-\frac{GMm}{r}$
Total	$-\frac{GMm}{2r}$

$$v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2g_0 R}$$

Escape speed does not depend on the mass of the object.

7 GEOSTATIONARY ORBIT

- Period 24 h, in the **equatorial plane**
- Moves west to east with the Earth: stays above one point
- Radius about 4.2×10^7 m (height ≈ 36 000 km)
- Astronauts feel weightless: they are in **free fall**, not free of gravity

8 COMMON SECTION A TRAPS

- Using the height h instead of $r = R + h$.
- Dropping the minus sign in V and U.
- Saying $g = 0$ in orbit.
- Thinking escape speed depends on the projectile mass.
- r doubled: F becomes $\frac{1}{4}$, V becomes $\frac{1}{2}$.

EXAM TIPS

- ✓Start satellite questions with "gravitational force provides the centripetal force".
- ✓Replace GM by $g_0 R^2$ when G and M are not given.
- ✓Use ratios for "how many times" questions.
- ✓Energy needed between orbits = ΔE_{total}.